

Full Stack + Data Science RAG Study Notes

This PDF is a structured knowledge base designed for Retrieval-Augmented Generation (RAG). It covers Full Stack Development and Data Science fundamentals for AI learning systems.

Data Science Overview

Data Science combines statistics, programming, and domain knowledge to extract insights from data.

Key Areas:

- Data Collection
- Data Cleaning
- Data Analysis
- Machine Learning
- Data Visualization

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Applications:

- Recommendation systems
- Fraud detection
- Predictive analytics

Statistics for Data Science

Core concepts:

- Mean, Median, Mode
- Standard deviation
- Probability distributions
- Hypothesis testing

Important ideas:

- Correlation vs causation
- Sampling methods

Machine Learning

Machine Learning is a subset of AI that enables systems to learn from data.

Types:

- Supervised Learning
- Unsupervised Learning

- Reinforcement Learning

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- KNN
- SVM
- Neural Networks

Workflow:

Data → Preprocessing → Training → Evaluation → Deployment

Full Stack Development

Full Stack Development includes frontend + backend + database.

Frontend:

- HTML
- CSS
- JavaScript
- React.js or Next.js

Backend:

- Node.js
- Express.js
- REST APIs

Database:

- MySQL
- PostgreSQL
- MongoDB

Concepts:

- Authentication
- JWT
- MVC architecture

Frontend Development

Frontend is the user interface layer.

Key concepts:

- DOM manipulation
- Responsive design
- Components
- State management

React basics:

- Components
- Props

- State
- Hooks

Backend Development

Backend handles server-side logic.

Key concepts:

- APIs (REST)
- Middleware
- Authentication
- Database integration

Node.js example use:

- Server creation
- Routing

Databases

Databases store structured data.

Types:

- SQL (Relational)
- NoSQL (Document-based)

Concepts:

- CRUD operations
- Indexing
- Joins

System Design Basics

System design focuses on scalable applications.

Topics:

- Load balancing
- Caching
- Microservices
- API gateways

Use cases:

- Netflix architecture
- Amazon systems